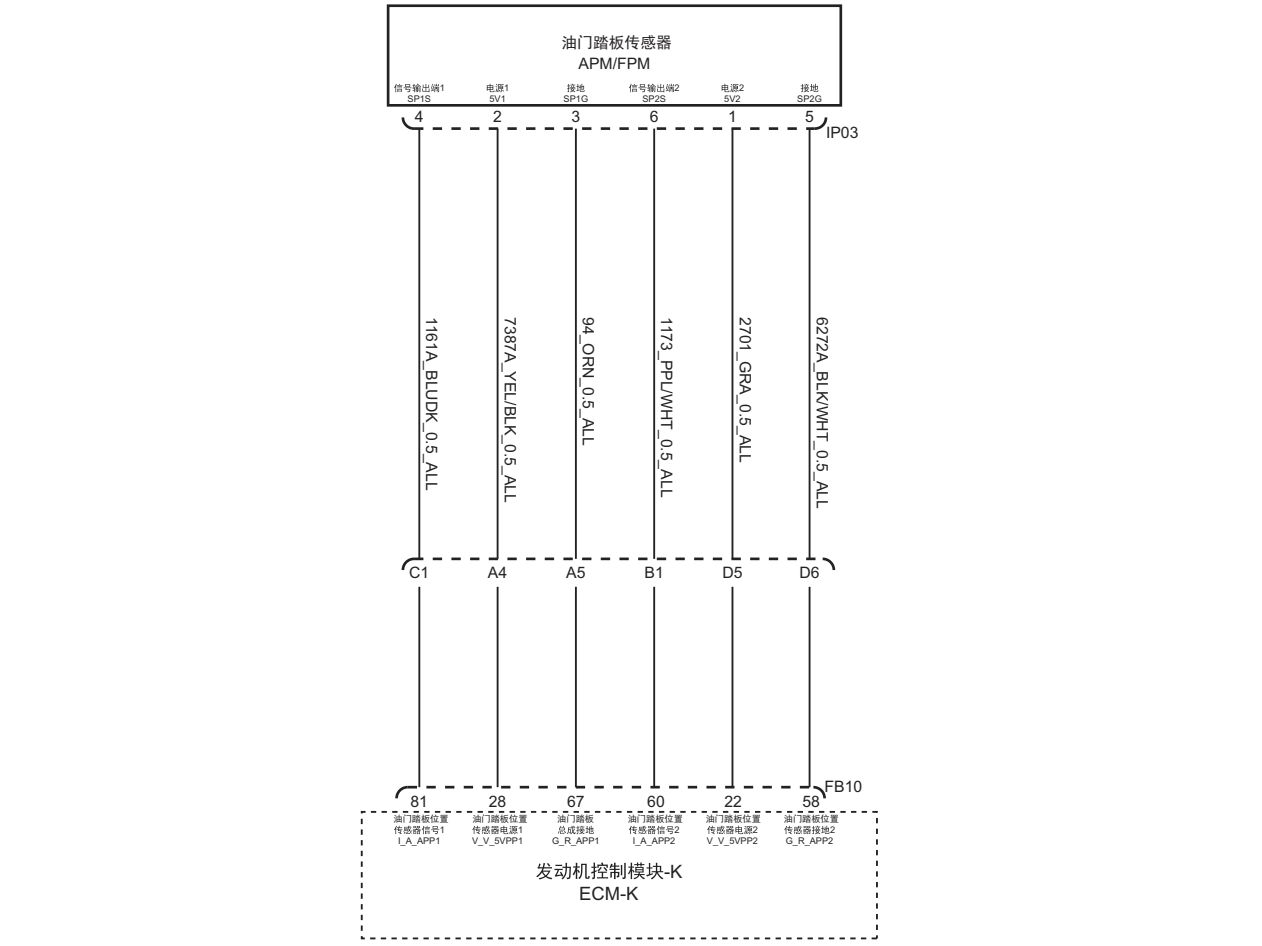
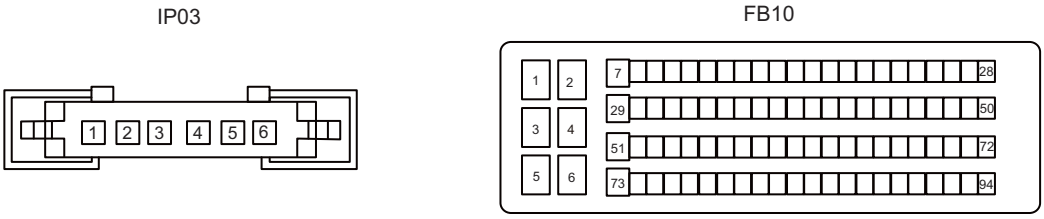
P0122、P0123、 P0222、 P0223、 P2135

Description of fault diagnosis codes
P0122-11：Signal Range Check Low for APP1
P0123-12：Signal Range Check High for APP1
P0222-11：Signal Range Check Low for APP2
P0223-12：Signal Range Check High for APP2
P2135-64：Plausibility Check between APP1 and APP2
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Fault in harness or harness plug • Accelerator pedal sensor error
Description of circuit diagram
There are two sensors of accelerator pedal position. The accelerator pedal position sensor 1 and 2 are installed above the assembly of accelerator pedal. Each sensor is equipped with one 5V reference voltage circuit, one reference voltage circuit of low level and one signal circuit. Both sensors are the potentiometer converting the accelerator pedal position into output voltage and then sending voltage signal to ECM. ECM will judge the idle position of accelerator pedal via the signal received by the accelerator pedal position sensor. ECM will operate the engine with such signals.

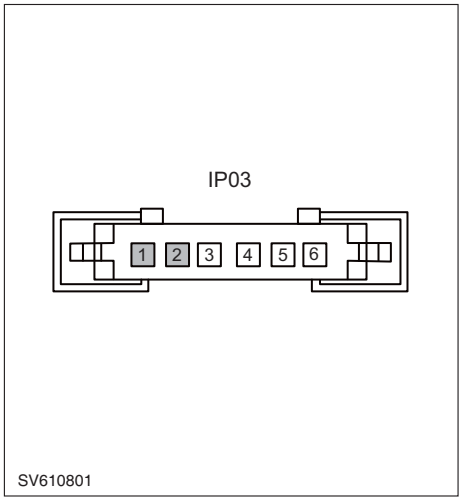
Circuit diagram

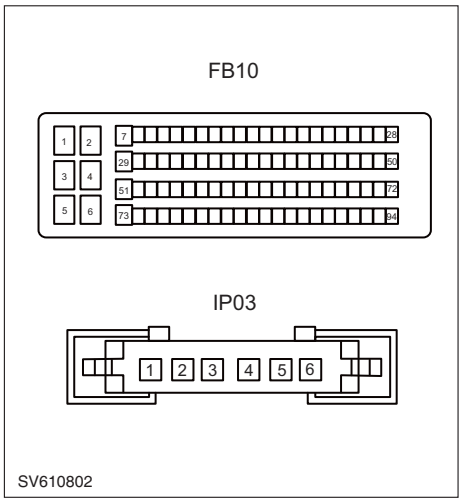


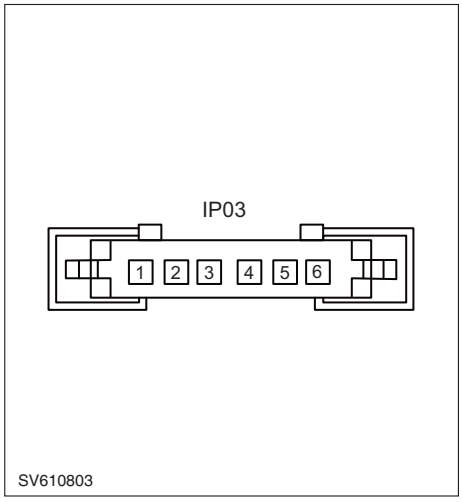


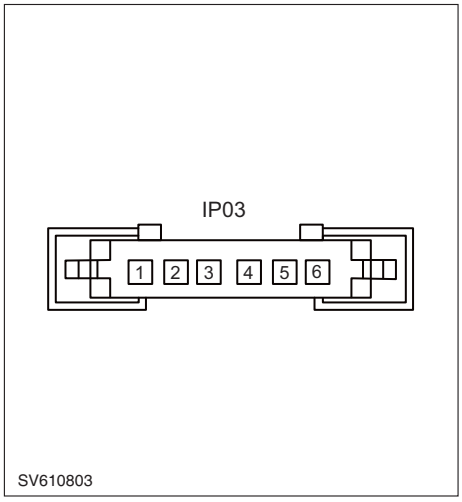
SV610800

Troubleshooting procedures		
Step 1	General inspection.	
		1 Check fuses for damages. 2 Check battery capacity. 3 Check the engine control module and the harness plug of accelerator pedal sensor for damage, poor contact, aging and looseness. 4 Check if results are normal?
		No → Repair or replace the faulted parts.
Yes ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the engine control system with diagnostic apparatus. 4 If there are other fault codes except for P0122, P0123, P0222, P0223 and P2135?
		Yes → Reference:DTC Summary List (EU5).
No ↓		
Step 3	Read the data flow of accelerator pedal sensor with diagnostic apparatus.	
		1 Connect the failure diagnosis instrument to the diagnosis interface. 2 Read the actual value of accelerator pedal sensor signal with diagnostic apparatus. **Standard value:** **Accelerator pedal opening: 0% (released); 100% (applied)** **Accelerator pedal sensor 1: 0.75V (released); 3.75V (applied)** **Accelerator pedal sensor 2: 0.37V (released); 1.84V (applied)** 3 Read whether the data flow is normal?
		Yes → Check is completed and confirm the system is operating normally.
No ↓		

Step 4	Check the power supply circuit of accelerator pedal sensor.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of accelerator pedal sensor. 3 Turn the ignition switch to ON position. 4 Measure the voltage between the harness plug of accelerator pedal sensor and the ground. Measure circuit: voltage between terminal 2 of IP03 plug and ground. Measure circuit: voltage between terminal 1 of IP03 plug and ground. **Standard value: 4.8~5.2V** 5 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

Step 5	Check the circuit of accelerator pedal sensor (open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of engine control module. 3 Disconnect the harness plug of accelerator pedal sensor. 4 Measure the resistance between the harness plug of engine control module and the harness plug of accelerator pedal sensor. **Accelerator pedal sensor 1:** Measure circuit: resistance between terminal 28 of FB10 plug and terminal 2 of IP03 plug. Measure circuit: resistance between terminal 81 of FB10 plug and terminal 4 of IP03 plug. Measure circuit: resistance between terminal 67 of FB10 plug and terminal 3 of IP03 plug. **Accelerator pedal sensor 2:** Measure circuit: resistance between terminal 22 of FB10 plug and terminal 1 of IP03 plug. Measure circuit: resistance between terminal 60 of FB10 plug and terminal 6 of IP03 plug. Measure circuit: resistance between terminal 58 of FB10 plug and terminal 5 of IP03 plug. **Standard value: ≈ 0Ω** 5 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

Step 6	Check the circuit of accelerator pedal sensor (short circuit to ground).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of engine control module. 3 Disconnect the harness plug of accelerator pedal sensor. 4 Measure the resistance between the harness plug terminal of accelerator pedal sensor and the ground. **Accelerator pedal sensor 1:** Measure circuit: resistance between terminal 2 of IP03 plug and ground. Measure circuit: resistance between terminal 4 of IP03 plug and ground. Measure circuit: resistance between terminal 3 of IP03 plug and ground. **Accelerator pedal sensor 2:** Measure circuit: resistance between terminal 1 of IP03 plug and ground. Measure circuit: resistance between terminal 6 of IP03 plug and ground. Measure circuit: resistance between terminal 5 of IP03 plug and ground. **Standard value: ∞** 5 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

Step 7	Check the circuit of accelerator pedal sensor (short circuit to battery).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of engine control module. 3 Disconnect the harness plug of accelerator pedal sensor. 4 Measure the voltage between the harness plug terminal of accelerator pedal sensor and the ground. **Accelerator pedal sensor 1:** Measure circuit: Voltage between terminal 2 of IP03 plug and the ground. Measure circuit: Voltage between terminal 4 of IP03 plug and the ground. Measure circuit: Voltage between terminal 3 of IP03 plug and the ground. **Accelerator pedal sensor 2:** Measure circuit: Voltage between terminal 1 of IP03 plug and the ground. Measure circuit: Voltage between terminal 6 of IP03 plug and the ground. Measure circuit: Voltage between terminal 5 of IP03 plug and the ground. **Standard value: ≈ 0V** 5 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

Step 8	Replace the accelerator pedal sensor.
	1 Replace the accelerator pedal sensor.
Next ↓	
Step 9	Complete the check and confirm whether the system works normally nor not.